

ICT702 Data Visualisation

Purposeful Use of Colour

Module 4



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.

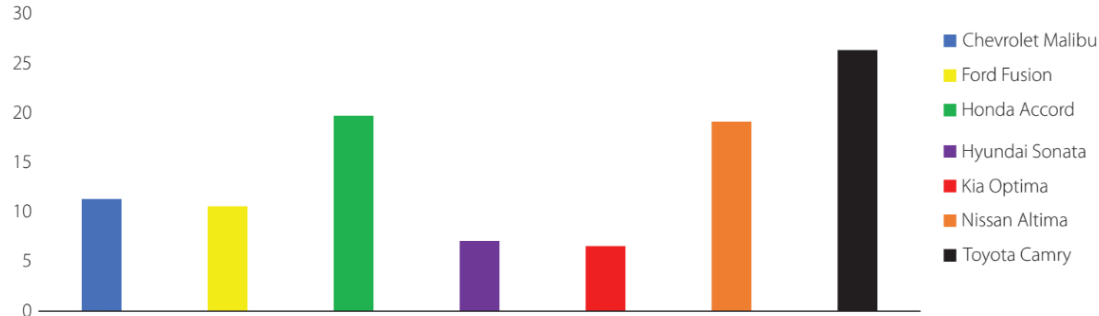


Part 3

Use of hue and a legend to identify categories

US Unit Sales for Top Selling Midsize Sedans

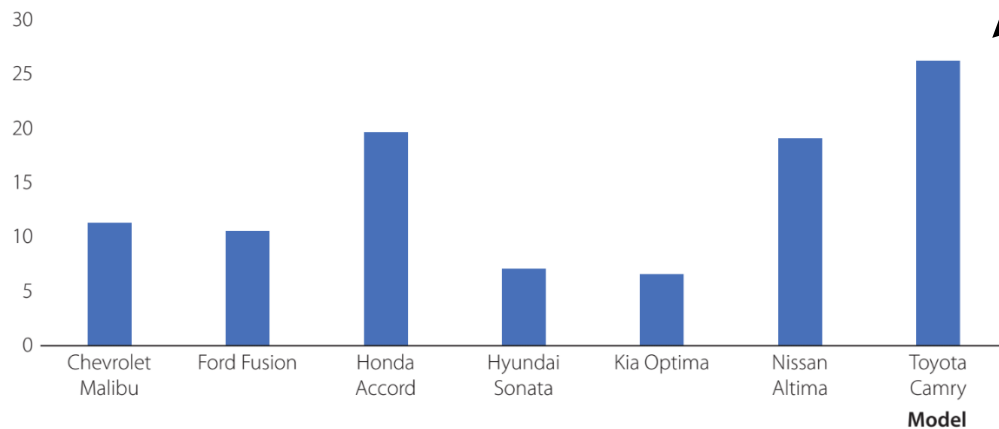
Sales (1000s)



Use of horizontal axis labels to identify categories

US Unit Sales for Top Selling Midsize Sedans

Sales (1000s)



Unnecessary Colour

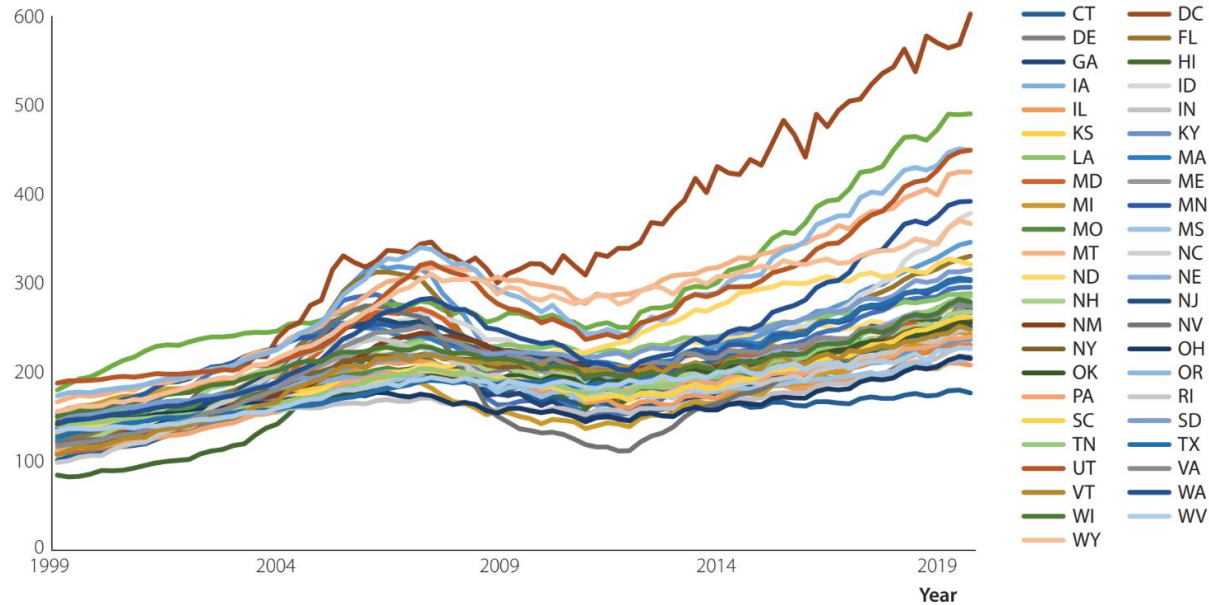
- A single colour replaces the colourful palette.
- The horizontal axis label replaces the legend with category names.
- The audience can now follow the chart from left to right with a reduced cognitive load.

Excessive Colour

Line chart of the quarterly house price index by state with too many colours

Quarterly House Price Index by State (Base = 1991)

Home Price Index



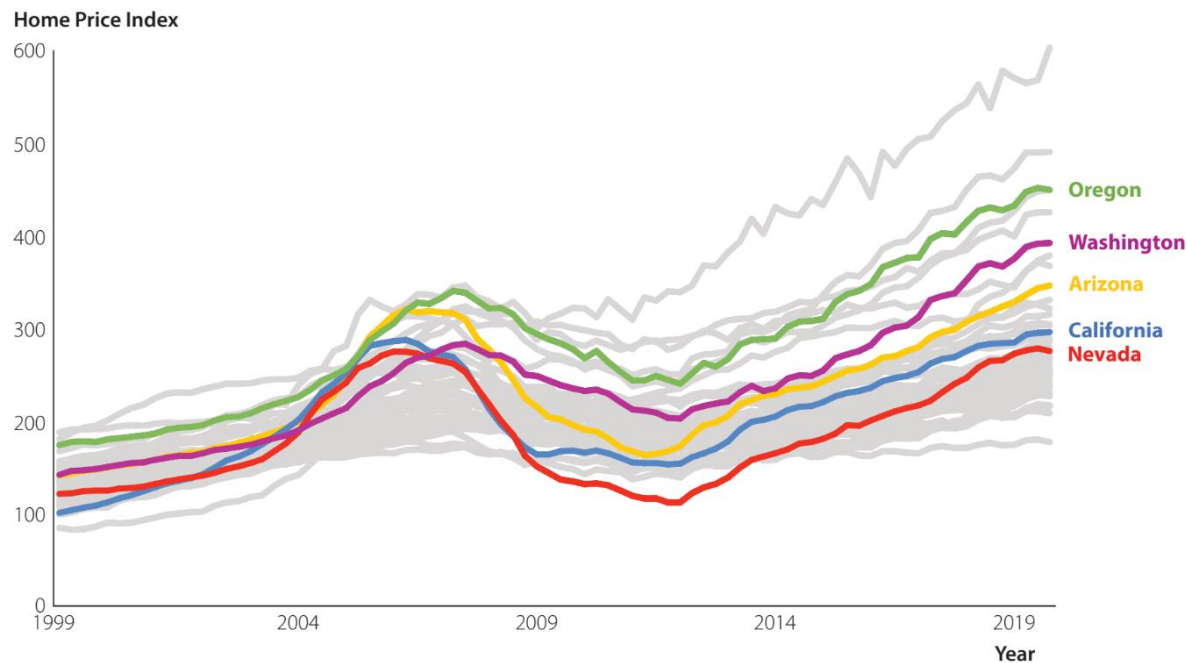
(data from <https://www.fhfa.gov/DataTools/Downloads/Pages/House-Price-Index-Datasets.aspx>)

With 51 categories/colours on this chart, comparing housing prices across states is difficult.

Reducing Excessive Colour

Chart of quarterly house price index by state with more efficient use of colour

Quarterly House Price Index (Base = 1991) for Western States Demonstrates More Volatility

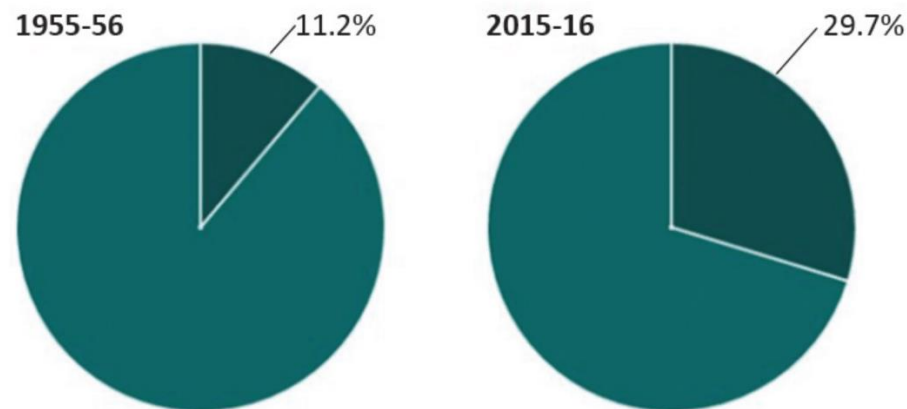


- Apply a different colour for each of the western states and use gray for all other states.
- Remove the legend and add a label for each of the five westernmost states to the end of their respective lines.

Insufficient Contrast

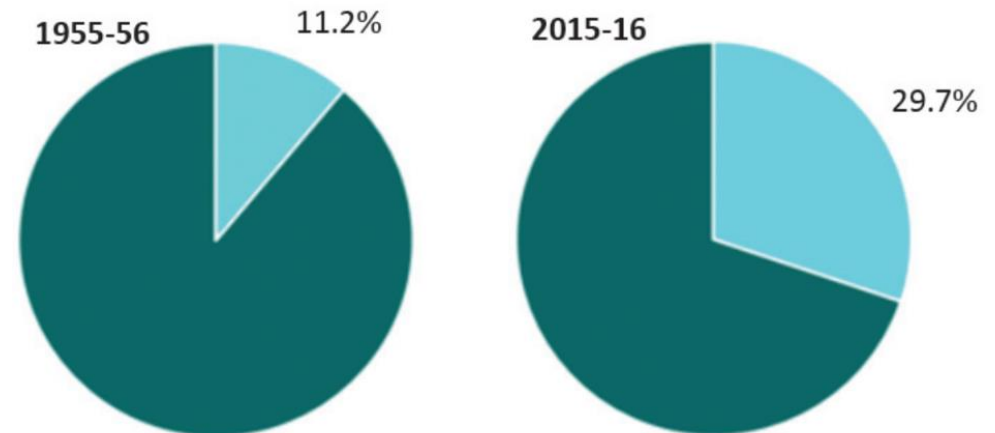
Lack of contrast makes it more challenging to differentiate between the pie chart slices

Change in proportion of public services budget spent on health
■ NHS ■ Rest of Budget



Increased contrast makes it easier to differentiate between the pie chart slices

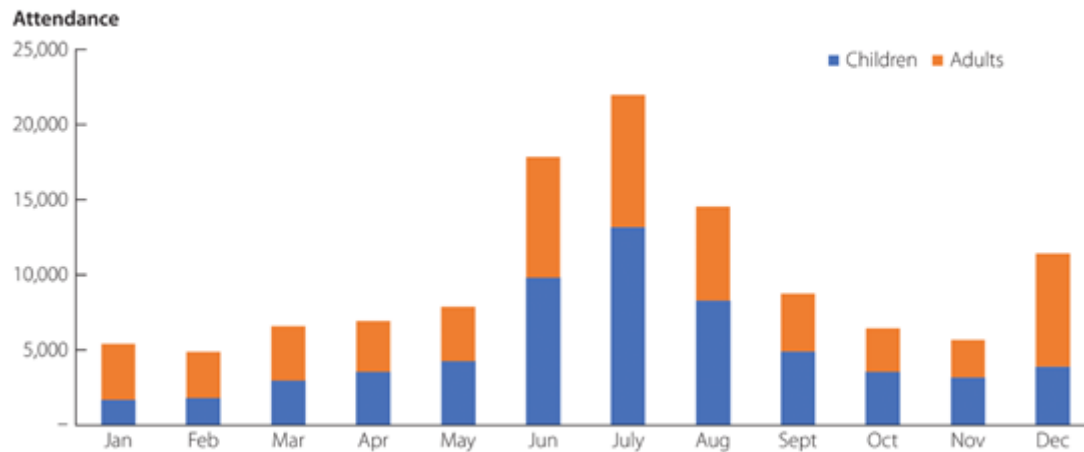
Change in proportion of public services budget spent on health
■ NHS ■ Rest of Budget



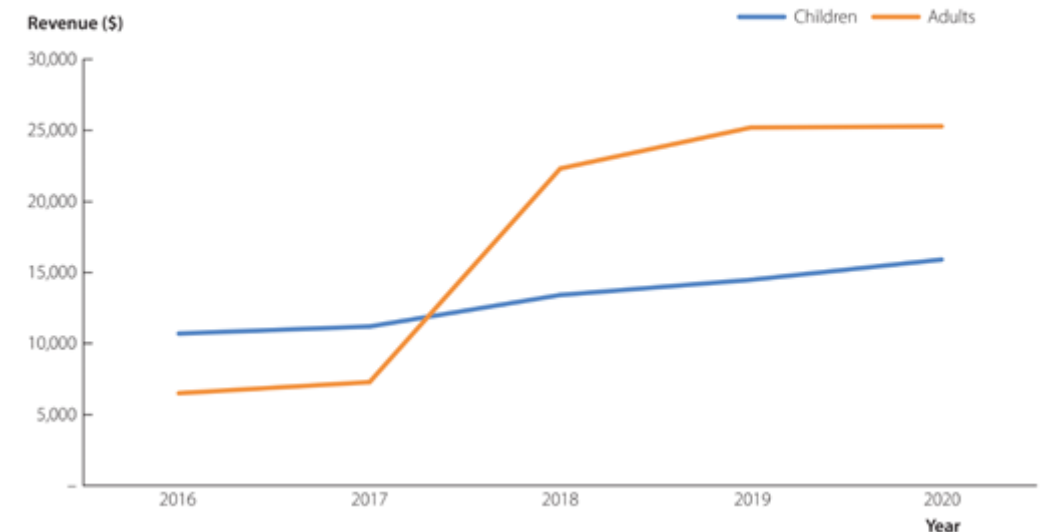
The increase in luminance difference increases the contrast between the two slices of each pie chart.

Inconsistency Across Charts

A stacked column chart of children and adult zoo attendance for the last year



A line chart of ticket revenue for December across several years



The consistent use of colours, orange and blue for adults and children, helps the audience comprehend the data presented across a series of related slides.

Neglecting Colourblindness

Colourblindness reduces the ability to accurately perceive some colours. The most common form of colourblindness is red-green colourblindness.

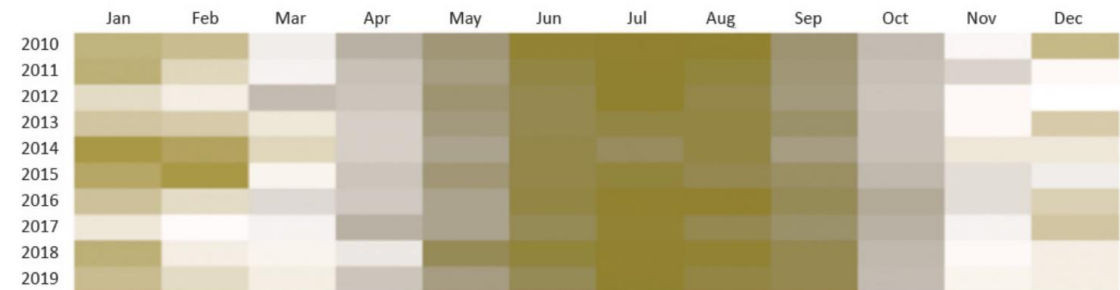
Heat map of monthly mean daily low temperature for Indianapolis 2010–2011

Monthly Mean Daily Low Temperature for Indianapolis



Heat map as It may appear to someone who is red-green colorblind

Monthly Mean Daily Low Temperature for Indianapolis



Neglecting Mode of Delivery

- Colour works differently in print and projection.
- Projected presentations should:
 - Use thick lines, sharp outlines, colour contrast, and relatively high saturation and luminance.
- Printed presentations should:
 - Use softer outlines with less saturated colours, lower luminance, and less contrast.

Remember colours vary across projectors and across printers so it's a good idea to review the output ahead of time.

Summary

In this module you have learnt:

- to use colour to define the pre-attentive attributes of hue, saturation, and luminance
- to distinguish between colour psychology and colour symbolism and use them effectively
- to design colour schemes that are appropriate for different variable types with meaningful reference value
- to use the HSL system for defining colour in Excel
- the common mistakes made when using colour

End of Part 3

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